F26 AerE 3310: Flight Control Systems I

Practice Problem Set 8: Frequency Response

Instructor: Abhishek Halder

All rights reserved.

Problem 1

(a) Compute $\angle z$ for z real. Compute the same for z pure imaginary.

Solution for Problem 1(a)

In general, for complex $z = \sigma + j\omega$, its real and imaginary parts $\sigma, \omega \in \mathbb{R}$. For z real, i.e., for $z = \sigma$, the definition of phase angle gives

$$\angle \sigma = \begin{cases} 0 & \text{if} \quad \sigma > 0, \\ \pi & \text{if} \quad \sigma < 0, \\ \text{undefined} & \text{if} \quad \sigma = 0. \end{cases}$$

For z pure imaginary, i.e., for $z = j\omega$, again the definition of phase angle directly gives

$$\angle j\omega = \begin{cases} \frac{\pi}{2} & \text{if} \quad \omega > 0, \\ -\frac{\pi}{2} \text{ (equivalently } \frac{3\pi}{2}) & \text{if} \quad \omega < 0, \\ \text{undefined} & \text{if} \quad \omega = 0. \end{cases}$$

(b) Compute $\angle f(z)$ in terms of z for

$$f(z) = -z,\ \bar{z},\ e^z,\ e^{-z},\ \log z,\ \frac{1}{z},\ z^p,$$

where $\bar{z}$ denotes the complex conjugate of z, and p is an integer.

Solution for Problem 1(b)

To avoid triviality, we consider z to be a nonzero complex number. For $f(z) = -z = (-1)z$, we use that the phase of product equals to the sum of phases. Hence

$$\angle(-z) = \angle(-1) + \angle z = \pi + \angle z.$$

For $f(z) = \bar{z}$, we take phase angle to both sides of $z\bar{z} = |z|^2 > 0$, to get $\angle z + \angle \bar{z} = 0$ using our answer from Prob. 1(a). Therefore,

$$\angle \bar{z} = -\angle z.$$

For $f(z) = e^z$, notice that if $z = \sigma + j\omega$, then

$$\angle e^z = \angle \left(e^{\sigma} \cdot e^{j\omega}\right) = \angle e^{\sigma} + \angle e^{j\omega} = 0 + \omega = \mathrm{Im}(z).$$

For $f(z) = e^{-z}$, notice that if $z = \sigma + j\omega$, then

$$\angle e^{-z} = \angle \left(e^{-\sigma} \cdot e^{-j\omega}\right) = \angle e^{-\sigma} + \angle e^{-j\omega} = 0 - \omega = -\,\mathrm{Im}(z).$$

For $f(z) = \log z$, notice that if $z = |z|e^{j\theta}$, then

$$\angle \log z = \angle \left(\log |z| + j\theta\right) = \theta + 2\pi r = (\angle z) + 2\pi k, \quad \text{for integer } k.$$

For $f(z) = 1/z$, we have

$$\angle \frac{1}{z} = \angle 1 - \angle z = 0 - \angle z = -\angle z.$$

For $f(z) = z^p$ with any integer p, and $z = |z|e^{j\theta}$, we have

$$\angle\, z^p = \angle |z|^p e^{jp\theta} = \angle e^{jp\theta} = p\angle z,$$

which generalizes the previous answer corresponding to the special case $p = -1$.

(c) Compute the gain $M(\omega)$ and phase $\phi(\omega)$ as function of the frequency ω for the PID controller

$$K(s) = k_{\mathrm{P}} + \frac{k_{\mathrm{I}}}{s} + k_{\mathrm{D}} s.$$

Solution for Problem 1(c)

We find the gain

$$M(\omega) = |K(j\omega)| = \left| k_{\mathrm{P}} + \frac{k_{\mathrm{I}}}{j\omega} + k_{\mathrm{D}} j\omega \right| = \left| \frac{k_{\mathrm{P}} j\omega + k_{\mathrm{I}} - k_{\mathrm{D}}\omega^2}{j\omega} \right| = \frac{|k_{\mathrm{P}} j\omega + k_{\mathrm{I}} - k_{\mathrm{D}}\omega^2|}{|j\omega|}$$
$$= \frac{\sqrt{k_{\mathrm{P}}^2\omega^2 + \left(k_{\mathrm{I}} - k_{\mathrm{D}}\omega^2\right)^2}}{\omega},$$

and the phase

$$\phi(\omega) = \angle \frac{k_{\rm P} j\omega + k_{\rm I} - k_{\rm D}\omega^2}{j\omega} = \angle \left(k_{\rm P} j\omega + k_{\rm I} - k_{\rm D}\omega^2\right) - \angle j\omega = -\frac{\pi}{2} + \angle \left(k_{\rm P} j\omega + k_{\rm I} - k_{\rm D}\omega^2\right),$$

since $0 < \omega < \infty$. More explicitly, using Lec. 22 p. 4, the phase

$$\phi(\omega) = -\frac{\pi}{2} + \begin{cases} \arctan \dfrac{k_{\rm P}\omega}{k_{\rm I} - k_{\rm D}\omega^2} & \text{if} \quad k_{\rm P}\omega > 0, k_{\rm I} - k_{\rm D}\omega^2 > 0, \\ \pi + \arctan \dfrac{k_{\rm P}\omega}{k_{\rm I} - k_{\rm D}\omega^2} & \text{if} \quad k_{\rm P}\omega > 0, k_{\rm I} - k_{\rm D}\omega^2 < 0, \\ -\pi + \arctan \dfrac{k_{\rm P}\omega}{k_{\rm I} - k_{\rm D}\omega^2} & \text{if} \quad k_{\rm P}\omega < 0, k_{\rm I} - k_{\rm D}\omega^2 < 0, \\ \arctan \dfrac{k_{\rm P}\omega}{k_{\rm I} - k_{\rm D}\omega^2} & \text{if} \quad k_{\rm P}\omega < 0, k_{\rm I} - k_{\rm D}\omega^2 > 0. \end{cases}$$

Problem 2

Consider the input to output transfer function

$$\frac{Y}{U} = \frac{0.1s + 0.01}{(s+1)(0.1s+1)}.$$

(a) Without using Laplace transform, analytically, i.e., by hand, derive the output $y(t)$ when the above system is excited with input $u(t) = \sin(2t)$. Show all calculations.

Solution for Problem 2(a)

From Lec. 23 p. 7, the input $u(t) = \sin(\omega t)$ results in the output $y(t) = M(\omega)\sin(\omega t + \phi(\omega))$ where the magnitude

$$M(\omega) = \frac{|0.1j\omega + 0.01|}{|j\omega + 1| \cdot |0.1j\omega + 1|} = \frac{0.1\sqrt{100\omega^2 + 1}}{\sqrt{\omega^2 + 1}\sqrt{\omega^2 + 100}},$$

and the phase

$$\phi(\omega) = \arctan\left(\frac{0.1\omega}{0.01}\right) - \arctan\left(\frac{\omega}{1}\right) - \arctan\left(\frac{0.1\omega}{1}\right) = \arctan 10\omega - \arctan\omega - \arctan 0.1\omega.$$

For $\omega = 2$, we get

$$M(2) = \frac{0.1 \times \sqrt{401}}{\sqrt{5} \times \sqrt{104}} \approx 0.0878, \quad \phi(2) = \arctan 20 - \arctan 2 - \arctan 0.2 \approx 0.2163\,\text{rad}.$$

Therefore, $u(t) = \sin(2t)$ results in the output

$$y(t) = M(2)\sin(2t + \phi(2)) \approx 0.0878\sin(2t + 0.2163).$$

(b) State if the quoted sentence is TRUE or FALSE: "For the given transfer function, the input frequency that maximizes the magnitude is not the same as the one that maximizes the phase". Give calculations and reasoning to justify your answer.

Solution for Problem 1(b)

TRUE. To see this analytically, notice that the phase maximizer ω_{phasemax} satisfies

$$\begin{aligned} & \frac{\mathrm{d}}{\mathrm{d}\omega}\phi(\omega)\Big|_{\omega=\omega_{\text{phasemax}}} = 0 \\ \Rightarrow \quad & \frac{10}{1+100\omega^2} - \frac{1}{1+\omega^2} - \frac{10}{100+\omega^2}\Big|_{\omega=\omega_{\text{phasemax}}} = 0 \\ \Leftrightarrow \quad & 10\left(1+\omega^2\right)\left(100+\omega^2\right)\Big|_{\omega=\omega_{\text{phasemax}}} = 11\left(10+\omega^2\right)\left(1+100\omega^2\right)\Big|_{\omega=\omega_{\text{phasemax}}}, \end{aligned} \tag{1}$$

and the magnitude maximizer ω_{magmax} satisfies

$$\begin{aligned} & \frac{\mathrm{d}}{\mathrm{d}\omega}M(\omega)\Big|_{\omega=\omega_{\text{magmax}}} = 0 \\ \Rightarrow \quad & \frac{\omega_{\text{magmax}}\left(100\omega_{\text{magmax}}^4 + 2\omega_{\text{magmax}}^2 - 9899\right)}{\left(1+\omega_{\text{magmax}}^2\right)^{3/2}\left(100+\omega_{\text{magmax}}^2\right)\sqrt{100\omega_{\text{magmax}}^2+1}} = 0 \\ \Rightarrow \quad & \omega_{\text{magmax}}\left(100\omega_{\text{magmax}}^4 + 2\omega_{\text{magmax}}^2 - 9899\right) = 0. \end{aligned}$$

The only positive real root for ω_{magmax} is $\frac{1}{10}\sqrt{99\sqrt{101}-1}$, the other roots

$$\omega_{\text{magmax}} = 0, -\frac{1}{10}\sqrt{99\sqrt{101}-1}, \pm\frac{j}{10}\sqrt{99\sqrt{101}+1}$$

are not admissible (either zero, or negative, or complex). At the admissible root

$$\omega_{\text{magmax}} = \frac{1}{10}\sqrt{99\sqrt{101}-1} \Leftrightarrow 100\omega_{\text{magmax}}^2 = 99\sqrt{101}-1,$$

which does not satisfy the equality (1) upon substitution.

(c) Write a MATLAB code plotting the Bode plots for the given transfer function. Look up the commands `tf` and `bode` in MATLAB documentation. This code can help you verify your hand calculations in parts (a)-(b).

Solution for Problem 1(c)

Please find the MATLAB code `PS08p2c.m` posted in CANVAS Files section, folder: Practice Problems and Solutions. This code verifies the analytic calculations in parts (a)-(b).